

Spotter Load Rate Prediction Model Report

1. Executive Summary

This report outlines the development and validation of a machine learning model designed to predict spot freight rates for Spotter. Using historical shipment data, feature engineering, and a LightGBM regressor, the model achieves a strong evaluation Mean Absolute Error (MAE) of **~\$142.73**, accurately capturing spot market dynamics without overfitting.

2. Validation Strategy (Time-Based Split)

To evaluate model performance realistically and prevent data leakage, a strict **time-based temporal split** was implemented rather than random k-fold cross-validation:

- **Training Period:** Data prior to September 1, 2025 (~38,477 samples).
- **Validation Period:** Data from September 1, 2025 onwards (~9,523 samples).

Rationale: In real-world freight pricing, future prices cannot inform past rate models. Evaluating on future temporal data ensures the model generalizes well to unseen market conditions.

3. Feature Engineering & Model Architecture

Key features constructed to capture price signals include:

- **Lane Representation:** Combined origin and destination (pickup -> delivery) to isolate lane-specific pricing baselines.
- **Temporal Dynamics:** Extracted month, day, dayofweek, dayofyear, and is_weekend to capture day-of-week demand spikes and seasonal trends.
- **Physical Interactions:** Calculated weight_per_mile to represent shipment density constraints.
- **Model Choice:** **LightGBM Regressor** was selected for its native handling of categorical features (lane, equipment, pickup, delivery), rapid training speeds, and resilience against outliers in freight pricing.

4. Results & Output Verification

- **Internal Split Metrics:**
 - **MAE:** \$142.73
 - **RMSE:** \$643.34
- **Validation Predictions (validation_predictions.csv):** Generated and verified 12,000 predictions matching expected schema and row constraints.

- **December Fixed Lane Predictions (data/december_chart_inputs.csv):**
Generated 31 daily rate forecasts for the fixed Lexington to Fort Wayne lane. The outputs were validated using score.py with 100% compliance

